PROJECT DOCUMENTATION REPORT

Full Stack & Automation Architecture — Streamlining IT Procurement

Date	1/10/2026	**Team ID**	SWTID-2026-4326
Project Title	Streamlining IT Procurement Automating Laptop Orders with Flow Designer		

1. INTRODUCTION

Project Title: Streamlining IT Procurement Automating Laptop Orders with Flow Designer

Team Members & Roles:

Name	Project Role	Responsibility
Racshana	**Team Lead**	Project Governance, Architecture Design & Flow Orchestration
Nivetha	**Team Member**	Frontend Portal Development & REST IntegrationHub Spokes
Yuvasri J	Team Member	AI/ML Predictive Model Engineering & Data Pipeline
Yuvashri SP	Team Member	Backend API Business Logic & Database Schema
Chudarshana Sree	Team Member	Quality Assurance, Testing & Security Validation

2. PROJECT OVERVIEW

Purpose: Modernize enterprise IT hardware ordering by replacing slow, email-driven procurement with a fully automated, transparent workflow powered by ServiceNow Flow Designer and AI recommendation engines.

Key Features: Role-based laptop catalog recommendations, real-time MS Teams approval dispatches, automated Workday HRIS hierarchy verification, vendor ERP order generation (Dell REST API), and end-to-end shipment tracking.

3. ARCHITECTURE

Frontend Architecture
Built on ServiceNow Service Portal using AngularJS, HTML5, and Tailwind CSS components. Delivers responsive laptop request forms with real-time field validation.

Backend Architecture
ServiceNow Flow Designer engine with Node.js/Express.js microservices for external integrations. Manages sub-flow orchestration, approvals, and error handlers.

Database & Data Layer
Relational CMDB & MongoDB store for asset tracking, request logs, and predictive ML model weights. Structured schema links users, departments, budget codes, and order statuses.

4. SETUP INSTRUCTIONS

Prerequisites: Node.js (v18+), MongoDB (v6.0+), ServiceNow Washington DC Developer Instance, REST API Client.

```
# Clone Repository & Setup Environment git clone https://github.com/swtid-2026-4326/it-procurement-flow-designer.git cd it-procurement-flow-designer npm install cp .env.example .env # Configure SNOW_INSTANCE_URL, API_KEY, MONGO_URI
```

5. FOLDER STRUCTURE

Client (Frontend)

```
client/ ├── src/ │ ├── components/ (CatalogCard, SpecSelector) │ ├── pages/ (OrderPortal, StatusTracker) │ └── services/ (snowApi.js)
```

Server (Backend & Flows)

```
server/ ├── flows/ (ProcurementMainFlow.xml) ├── routes/ (approvalRoutes.js, vendorApi.js) ├── models/ (AssetSchema.js, RequestLog.js) └── config/ (db.js, integrationHub.js)
```

6. RUNNING THE APPLICATION

Execute commands in separate terminal sessions for full-stack execution:

```
# Frontend Client (Port 3000) cd client && npm start # Backend Server & REST Spokes (Port 5000) cd server && npm start
```

7. API DOCUMENTATION

Method	Endpoint	Parameters / Body	Response
POST	/api/orders/submit	{ empId, laptopTier, budgetCode }	201 Created (Req ID)
POST	/api/approvals/teams	{ requestId, managerId, action }	200 OK (Status Updated)
GET	/api/vendor/dell/po/:id	requestId (URL param)	200 OK (PO Details)

8. AUTHENTICATION & 9. USER INTERFACE

OAuth 2.0 with JWT tokens and SAML Single Sign-On (SSO) integrated via ServiceNow Security Rules. The UI delivers an intuitive catalog with real-time status updates.

10. TESTING, 11. SCREENSHOTS & APPENDIX

Comprehensive Jest & ServiceNow Automated Test Framework (ATF) suites executed. Below is the primary UI banner design and workflow overview:



12. Known Issues

• Minor vendor REST API rate-limiting delays during peak fiscal year-end order bursts.

13. Future Enhancements

• Multi-vendor price comparison engine.
• Predictive replacement triggers prior to laptop hardware degradation.